

# GROUP 16

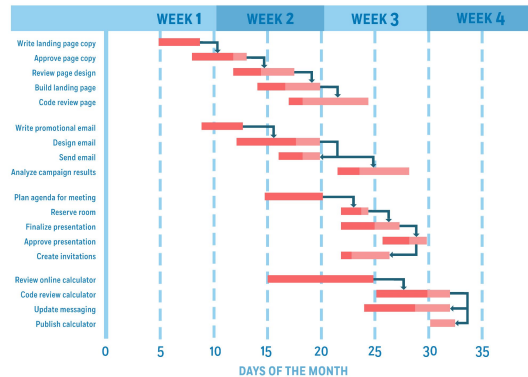
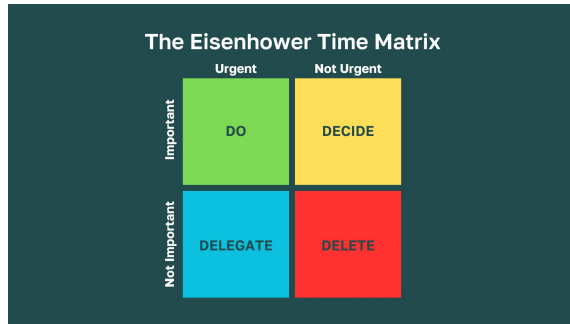
Alina Wang, Esme Rubinstein, Minsik Jeon, Shunxuan Wang, Itak Choi

# HeteroMRTA-Priority: Teaching Multi-Robot Systems to Eat The Frog 🐸

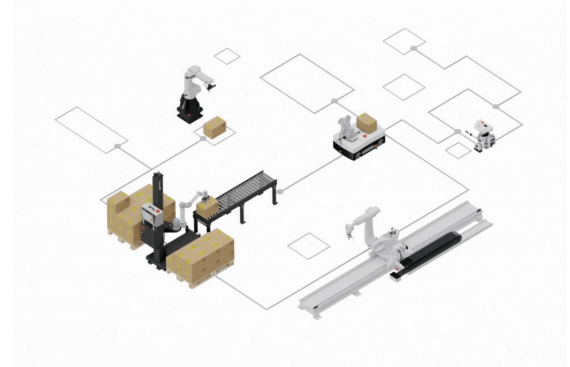
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# Background - Why does Heterogeneous Multi-Robot Task Allocation(MRTA) Matter?

## Human Team



## Multi-Robot System



## Previously on HeteroMRTA - Dai et. al, IEEE RAL 2025

### MRTA in General

Recent work is shifting toward **Reinforcement Learning** for task allocation.

- Can find near-optimal solution quickly → good for real-world applications
- Good scalability with task and robot complexity

### Bischoff et al. (2020) – Multi-Robot Task Allocation w/ Precedence Constraints

- Uses heuristic scheduling + precedence logic.
- Handles ordering well but not scalable and not RL-driven.

### Dai et al. (2025): Heterogeneous MRTA (Hetero-MRTA)

Dai's group focus on **ST-MR-TA** MRTA: Single-Task Robots, Multi-Robot Tasks, Time-Extended Assignment

- Modeled cooperative tasks requiring multiple robots with complementary traits
- Developed decentralized RL decision-making framework without requiring demonstration data (unlike imitation-learning method like [Jose & Zhang 2024](#))

# Problem Statement

## Goal of Dai's paper

- Minimize makespan with this reward function:
  - $R(\Phi) = -T - W$ , where  $T$  is the makespan and  $W$  represents the average ability redundancy for tasks

## Limitation of Dai's paper: **it optimizes for routing, but not for utility**

- Treats all tasks equally
  - Moving debris = Transporting patients
  - Mapping an unknown area = Locating a child in a collapsing building
- No priority or ordering logic



!?  
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# Project Idea - What happens when not all tasks are equal?

## We extend upon Dai's work:

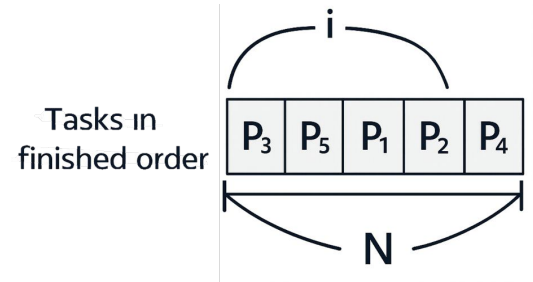
- Priority rating for different tasks in various environments
- Corresponding rewards function for completing high priority tasks
- Minimize makespan **while maximizing the timeliness of high-value tasks**

Final Reward Function:  $R(\Phi) = -T -W + P$

Priority Reward:

$$P = \omega \times \sum_i \frac{N-i}{N} p_i$$

$p_i$  : task priority,  $N$  : total tasks,  $i$  : finished order,  $\omega$  : weight

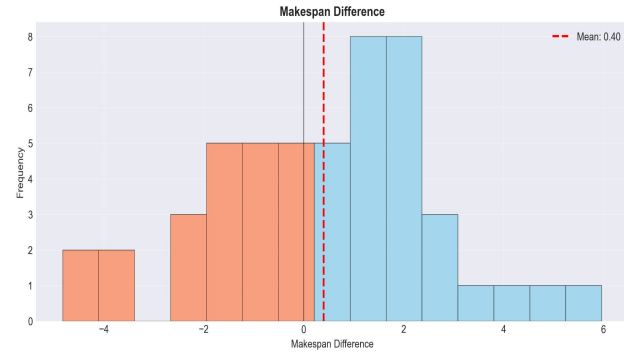


## Weight to Priority Rewards:

- **w = 0.5:** Priority matters a lot, robots aggressively serve high-value tasks first
- **w = 0.1:** Priority matters a little, robots stay mostly efficient but slightly favor important tasks
- **w = 0:** No priority, robots optimize purely for makespan and travel time (Baseline)

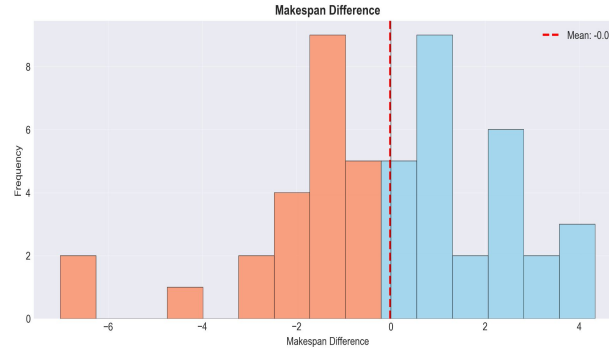
# Results - Makespan

## No Priority vs 0.1 Priority



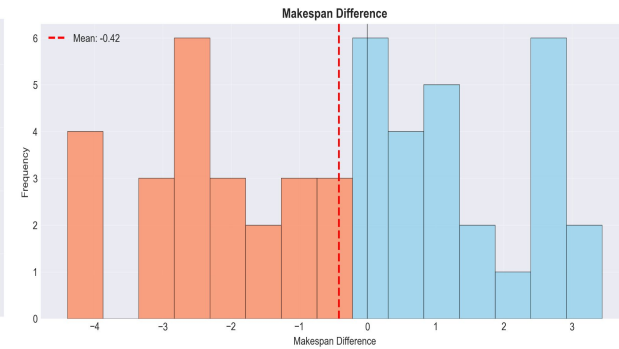
**Makespan: 0.40 units reduction**

## No Priority vs 0.5 Priority



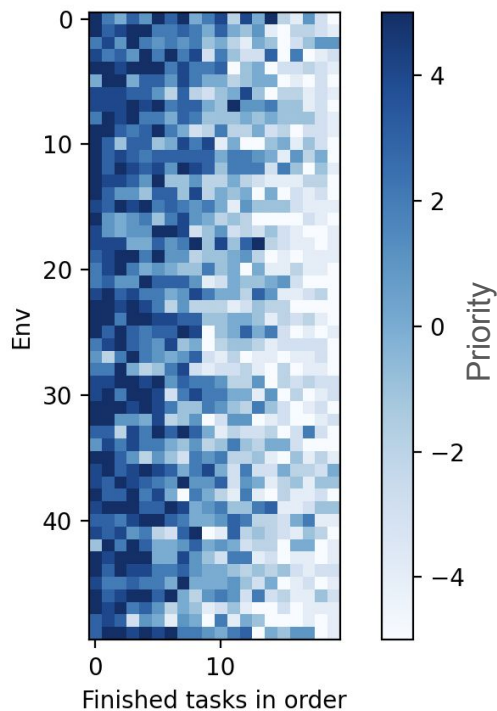
**Makespan: 0.02 units increase**

## 0.1 Priority vs 0.5 Priority

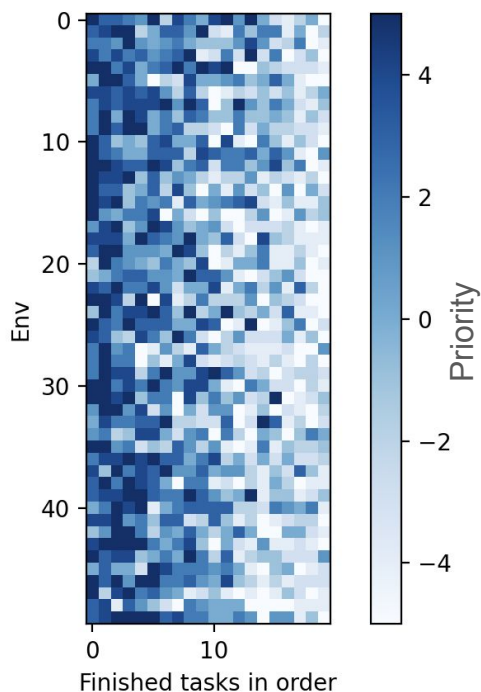


**Makespan: 0.42 units lower**

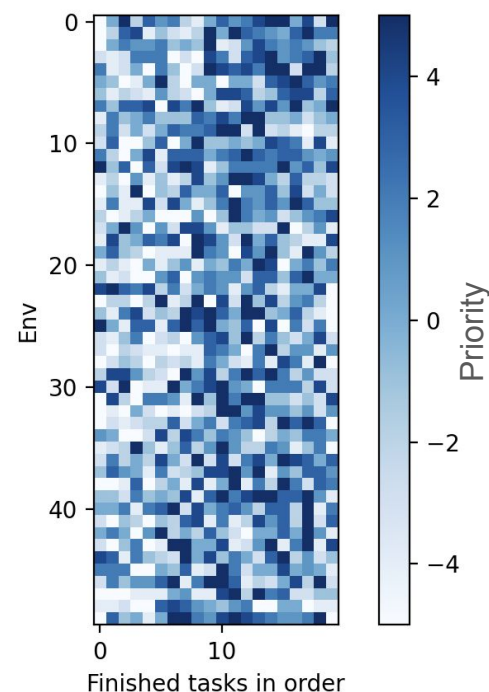
## Results - Visualization of Finished Task in Order



**Priority weight 0.5**



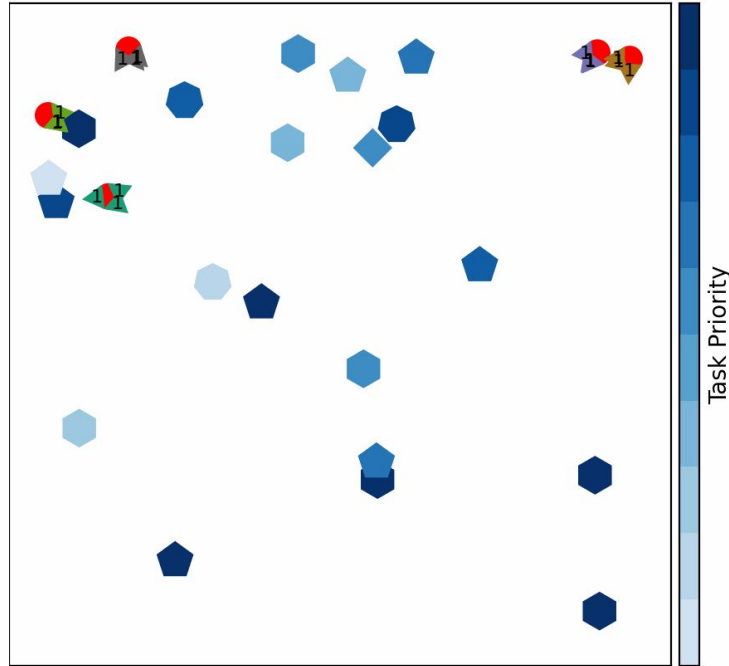
**Priority weight 0.1**



**Without priorities**

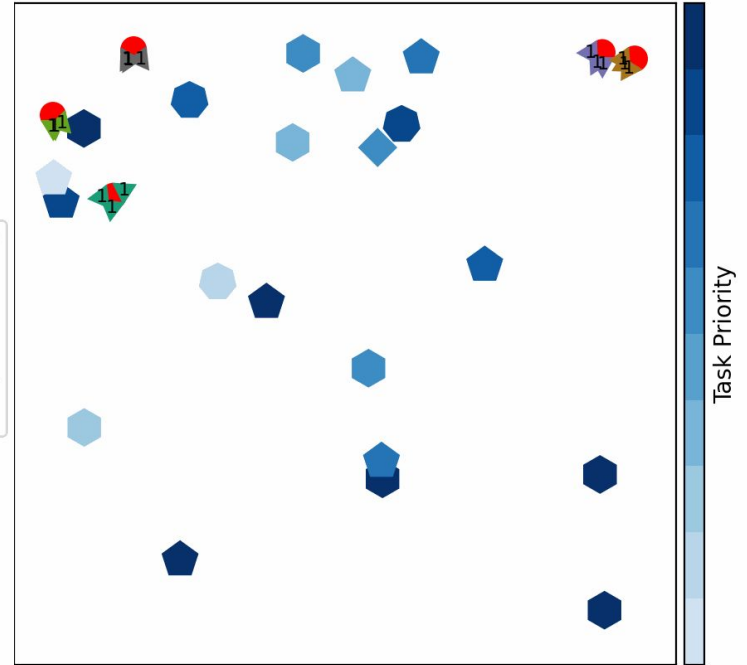
# Results

Agents finish 100.0% tasks within 20.38min.  
Current time is 0.00min



With priorities (-5 to 5)

Agents finish 100.0% tasks within 20.71min.  
Current time is 0.00min



Without priorities

- Agent species 0
- Agent species 1
- Agent species 2
- Agent species 3
- Agent species 4
- Finished task

## **Next Steps - More Realistics Deployment Considerations**

### **Explore Results**

- Further explore why makespan improved with priority weight 0.1
  - Try with more test sets to see behavior in more scenarios

### **Dynamic Priorities**

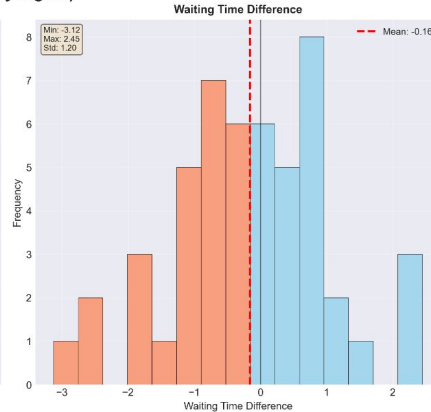
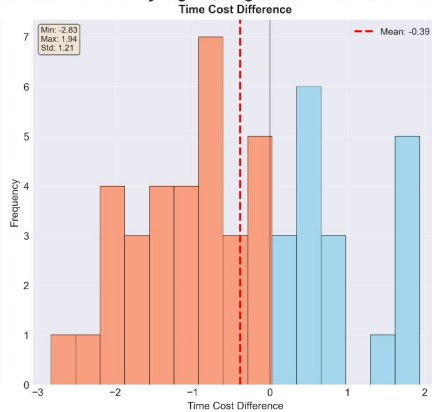
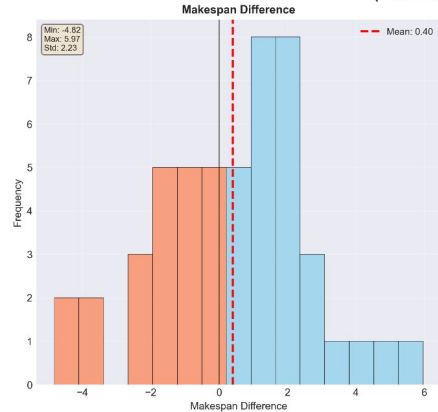
- Shift from static priority to time-dependent urgency
  - Connect to external API on weather, forest-fire warning, etc

### **Strict Constraints**

- Adding prerequisites dependencies
- Going beyond sequential execution requirements

# Results - No Priority vs. 0.1 Priority

Comparison: No Priority vs 0.1 Priority  
(Positive values = No Priority higher; Negative values = 0.1 Priority higher)

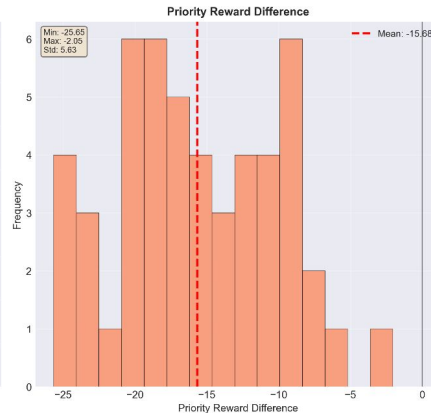
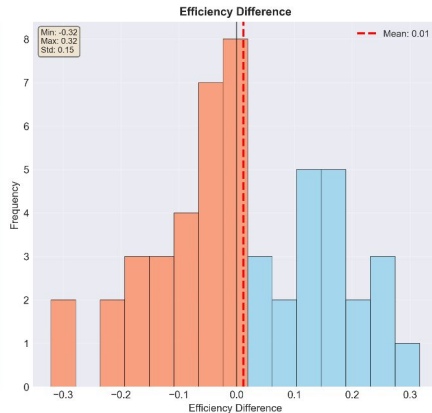
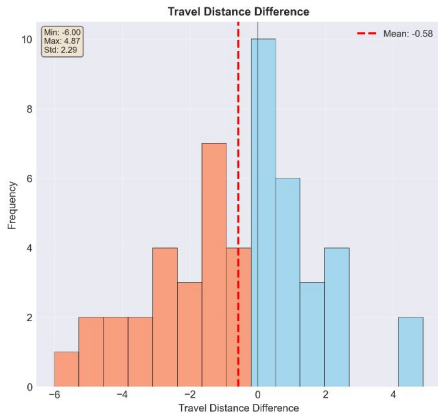


## 0.1 Priority System Advantages:

- **Makespan:** 0.40 units reduction
- **Priority:** 15.68 units higher

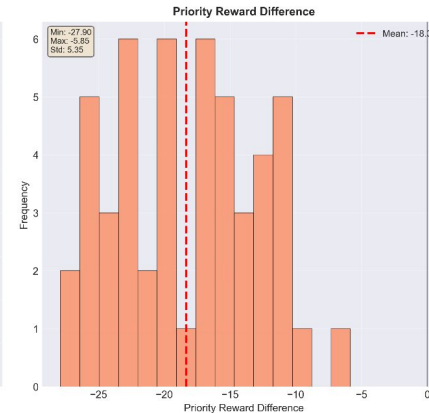
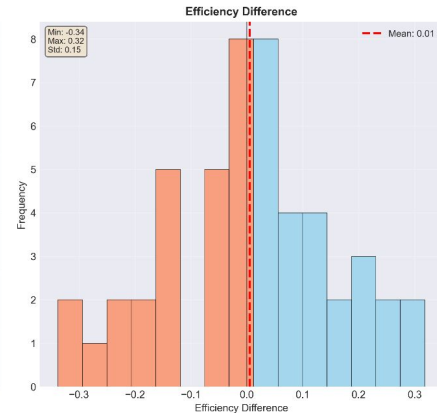
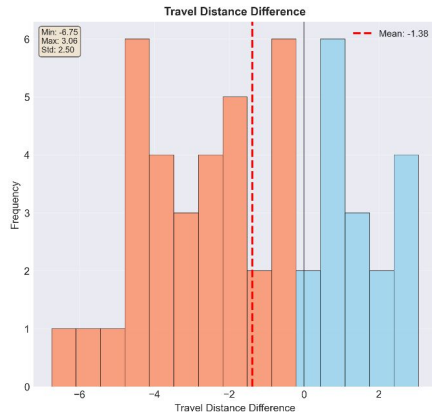
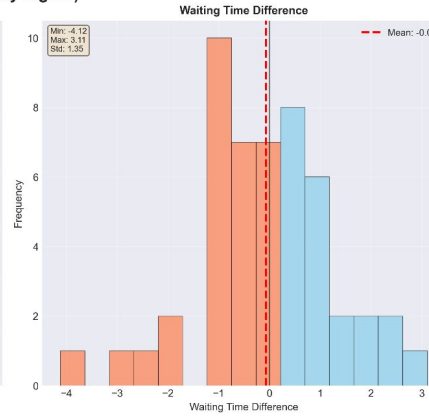
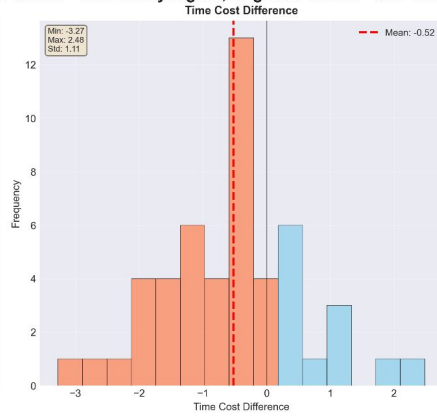
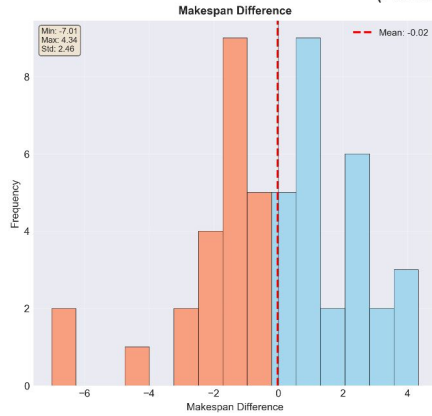
## 0.1 Priority System Disadvantages:

- **Time Cost:** 0.39 units higher
- **Travel Distance:** 0.58 units longer



# Results - No Priority vs. 0.5 Priority

Comparison: No Priority vs 0.5 Priority  
(Positive values = No Priority higher; Negative values = 0.5 Priority higher)



## 0.5 Priority System Advantages:

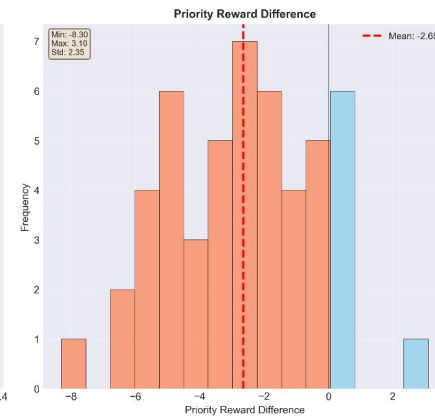
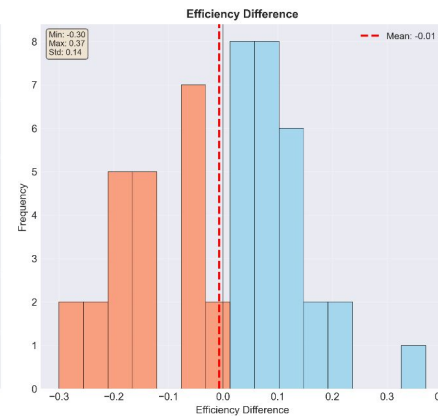
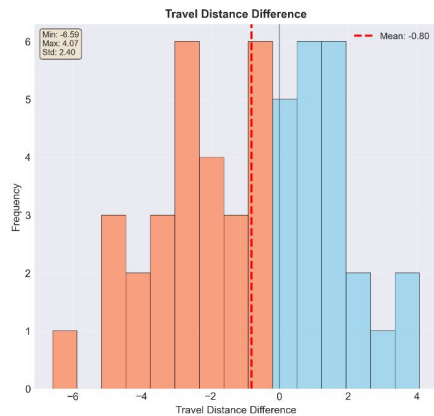
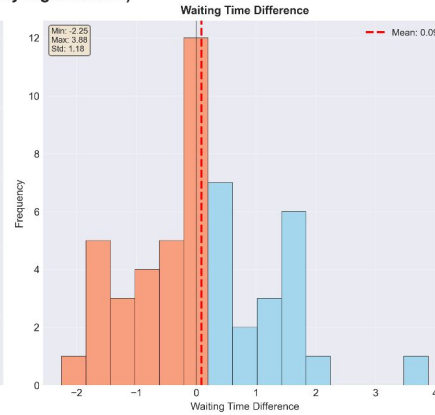
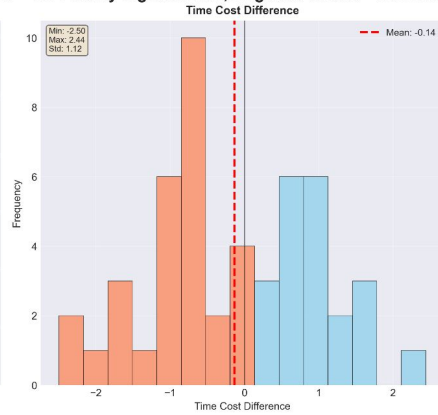
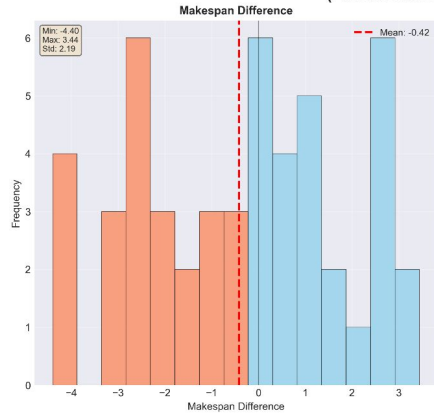
- **Priority:** 18.32 units higher

## 0.5 Priority System Disadvantages:

- **Makespan:** Marginal 0.02 units increase
- **Time Cost:** 0.52 units higher
- **Travel Distance:** 1.38 units longer
- **Waiting Time:** 0.07 units longer

# Results - 0.1 Priority vs. 0.5 Priority

Comparison: 0.1 Priority vs 0.5 Priority  
(Positive values = 0.1 Priority higher/better; Negative values = 0.5 Priority higher/better)



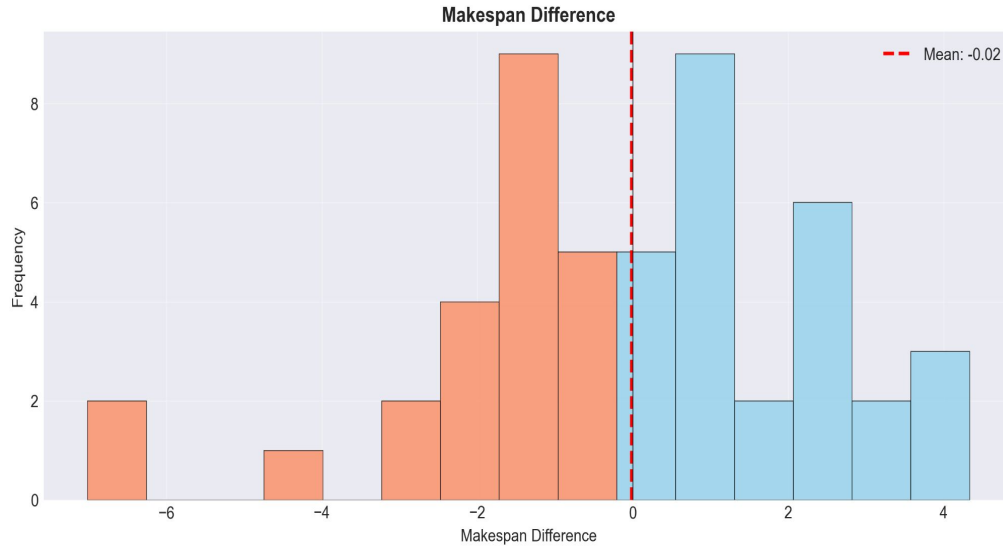
## 0.1 Priority System Advantages:

- **Makespan:** 0.42 units lower
- **Time Cost:** 0.14 units lower
- **Travel Distance:** 0.80 units lower

## 0.5 Priority System Advantages:

- **Priority:** 2.65 units higher

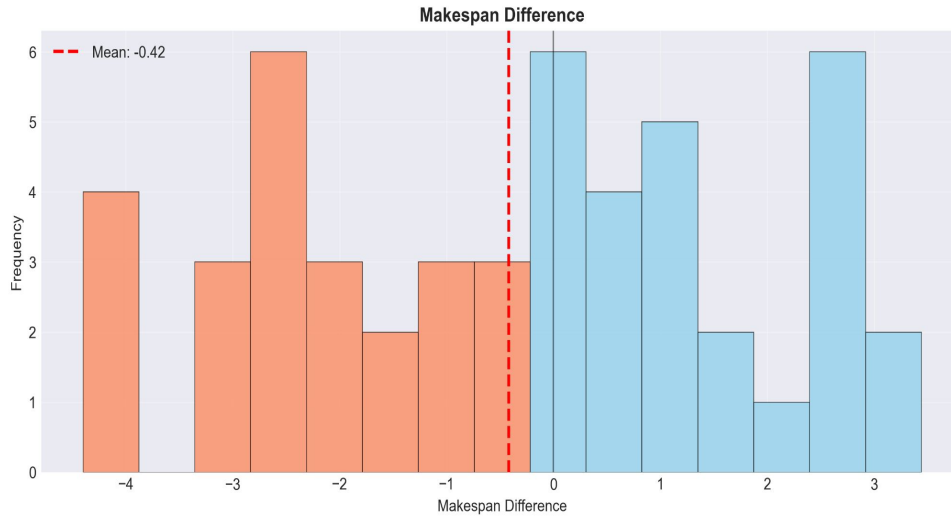
# Results - No Priority vs. 0.5 Priority



## 0.5 Priority System:

- **Priority:** 18.32 units higher
- **Makespan:** Marginal 0.02 units increase

# Results - 0.1 Priority vs. 0.5 Priority



## 0.1 Priority System:

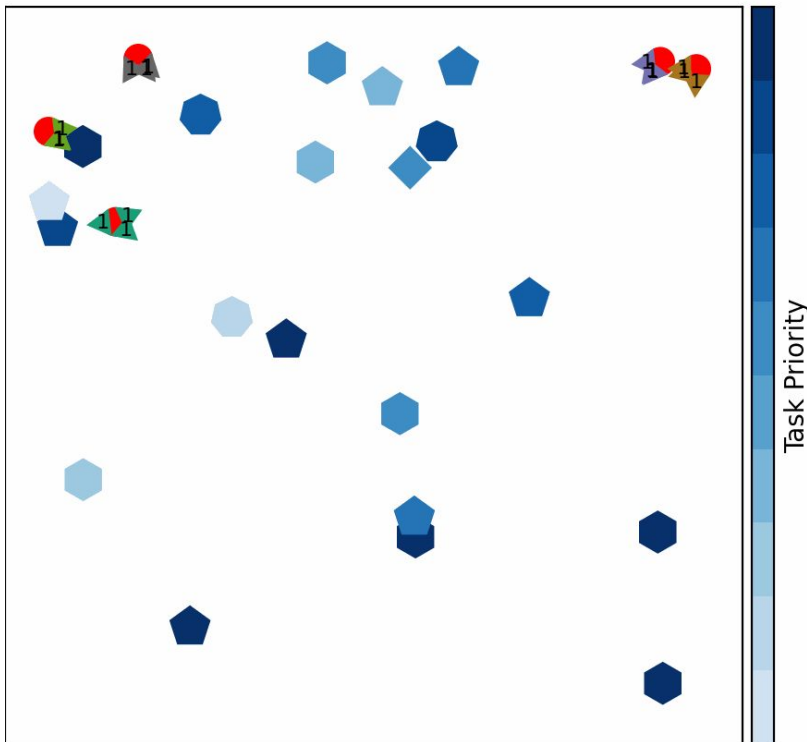
- **Makespan:** 0.42 units reduction

## 0.5 Priority System:

- **Priority:** 2.65 units higher

Agents finish 100.0% tasks within 20.38min.  
Current time is 0.00min

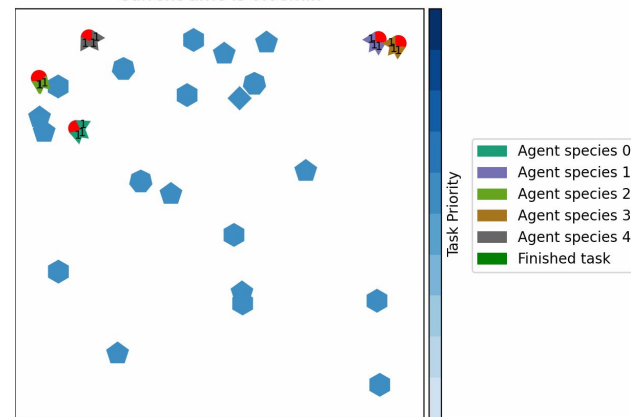
w/ priorities  
-5 to 5



- Agent species 0
- Agent species 1
- Agent species 2
- Agent species 3
- Agent species 4
- Finished task

Agents finish 100.0% tasks within 18.51min.  
Current time is 0.00min

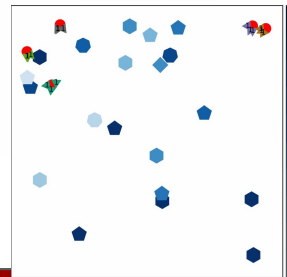
w/ 0 priority



- Agent species 0
- Agent species 1
- Agent species 2
- Agent species 3
- Agent species 4
- Finished task

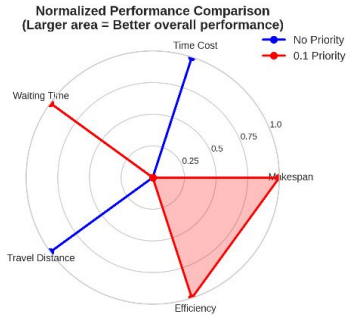
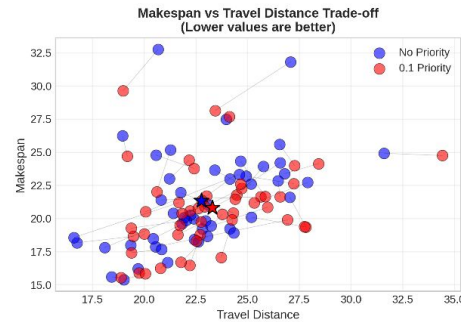
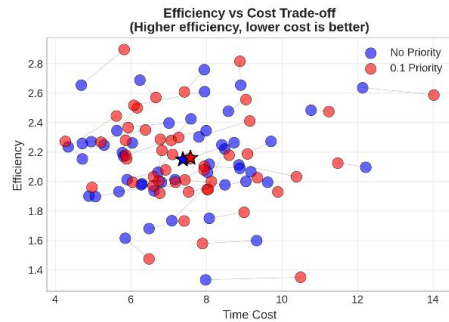
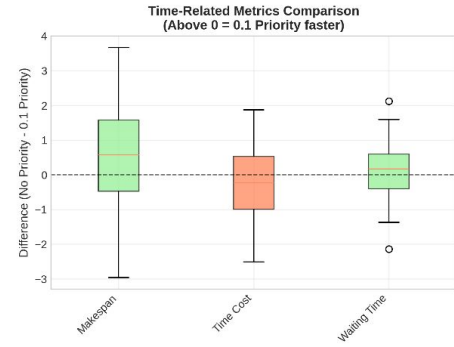
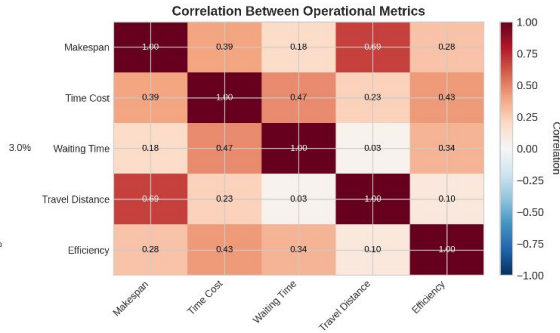
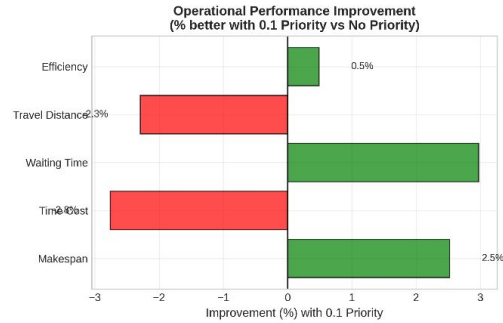
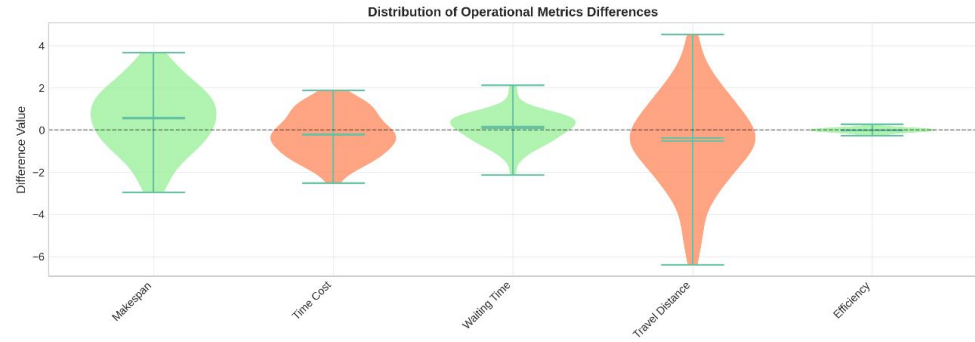
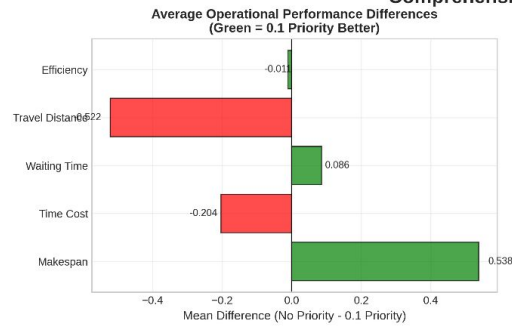
w/ priority but  
reward has no  
priority

Agents finish 100.0% tasks within 20.71min.  
Current time is 0.00min



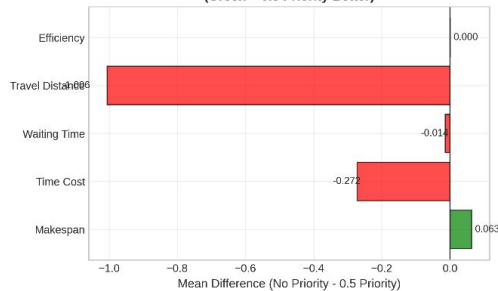
- Agent species 0
- Agent species 1
- Agent species 2
- Agent species 3
- Agent species 4
- Finished task

# Comprehensive Operational Performance Analysis: No Priority vs 0.1 Priority System

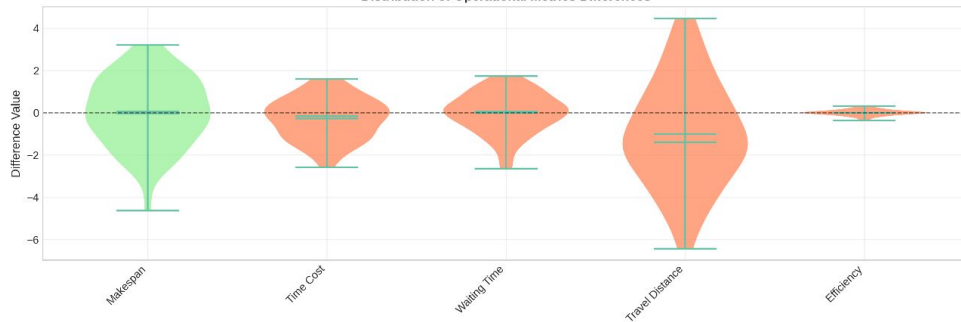


# Comprehensive Operational Performance Analysis: No Priority vs 0.5 Priority System

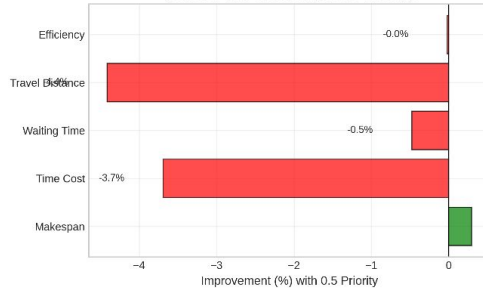
Average Operational Performance Differences  
(Green = 0.5 Priority Better)



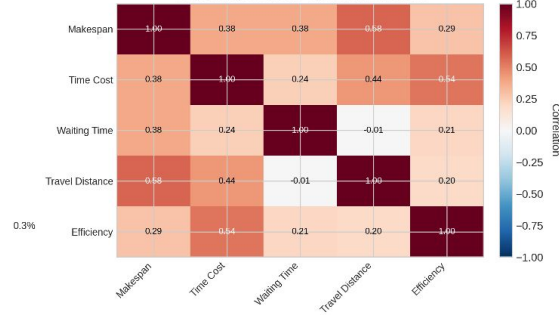
Distribution of Operational Metrics Differences



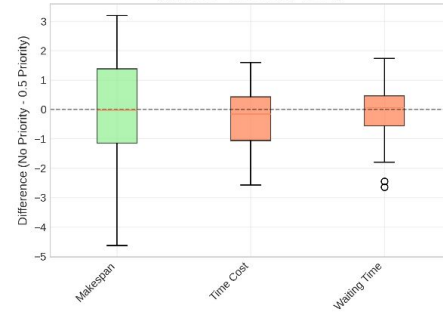
Operational Performance Improvement  
(% better with 0.5 Priority vs No Priority)



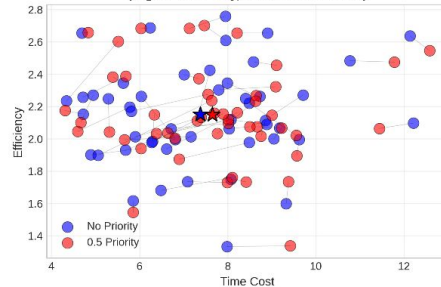
Correlation Between Operational Metrics



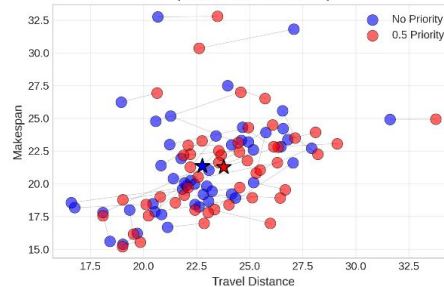
Time-Related Metrics Comparison  
(Above 0 = 0.5 Priority faster)



Efficiency vs Cost Trade-off  
(Higher efficiency, lower cost is better)



Makespan vs Travel Distance Trade-off  
(Lower values are better)



Normalized Performance Comparison  
(Larger area = Better overall performance)

